1.TIC-TAC-TOE

```
def print_board(board):
    print()
    for i in range(3):
        print(" ".join(board[i]))
    print()

def check_winner(board, player):
    for i in range(3):
        if all(board[i][j] == player for j in range(3)):
            return True
        if all(board[j][i] == player for j in range(3)):
            return True

    if all(board[i][i] == player for i in range(3)):
        return True

    if all(board[i][2 - i] == player for i in range(3)):
        return True

    return False

def minimax(board, is_max):
    if check_winner(board, "O"):
        return 1
    if check_winner(board, "X"):
        return -1
```

```
    if all(board[i][j] != "_" for i in range(3) for j in range(3)):
        return 0

    if is_max:
        best = -float("inf")
        for i in range(3):
            for j in range(3):
                if board[i][j] == "_":
                    board[i][j] = "O"
                    score = minimax(board, False)
                    board[i][j] = "_"
                    best = max(best, score)
        return best

    best = float("inf")
    for i in range(3):
        for j in range(3):
            if board[i][j] == "_":
                board[i][j] = "X"
                score = minimax(board, True)
                board[i][j] = "_"
                best = min(best, score)
    return best



import random

def computer_move(board):
    empty = [
```

```
        (i, j)
        for i in range(3)
        for j in range(3)
        if board[i][j] == "_"
    ]

    if empty:
        i, j = random.choice(empty)
        board[i][j] = "O"
        print(f"Computer places O at ({i + 1},{j + 1})")
def main():
    board = [["_"] * 3 for _ in range(3)]

    print("TIC-TAC-TOE")
    print("You are X. Computer is O.")
    print("Enter row and column numbers from 1 to 3.")

    while True:
        print_board(board)

        try:
            row, col = map(int, input("Your move (row column): ").split())
        except ValueError:
            print("Enter two numbers, for example: 1 2")
            continue

        row -= 1
        col -= 1
```

```
if not (0 <= row < 3 and 0 <= col < 3):
    print("Choose row and column between 1 and 3.")
    continue

if board[row][col] != "_":
    print("That position is occupied. Try again.")
    continue

board[row][col] = "X"

if check_winner(board, "X"):
    print_board(board)
    print("You Win!")
    break

if all(board[i][j] != "_" for i in range(3) for j in range(3)):
    print_board(board)
    print("Game Draw")
    break

computer_move(board)

if check_winner(board, "O"):
    print_board(board)
    print("Computer Wins!")
    break

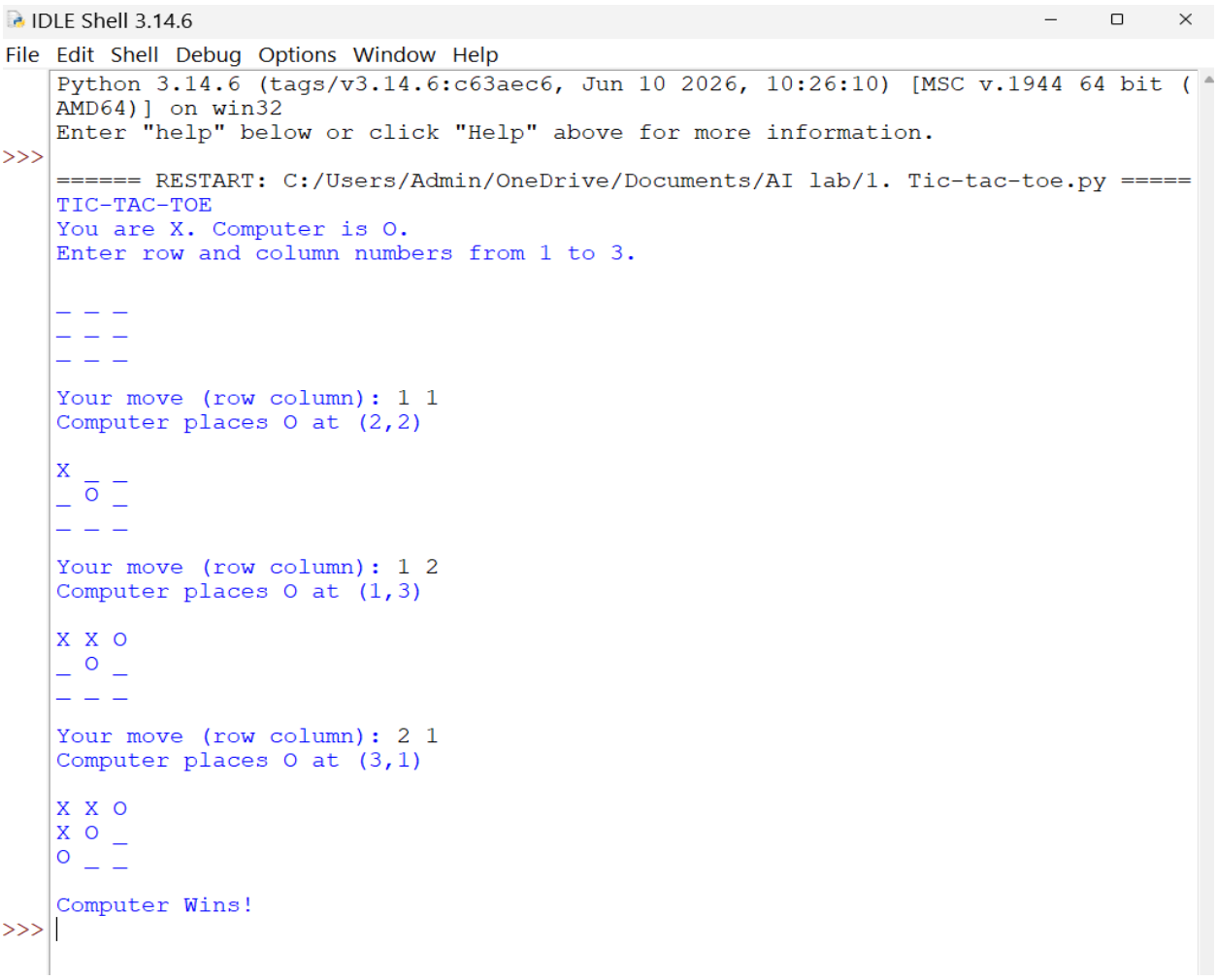
if all(board[i][j] != "_" for i in range(3) for j in range(3)):
    print_board(board)
```

```
        print("Game Draw")
        break

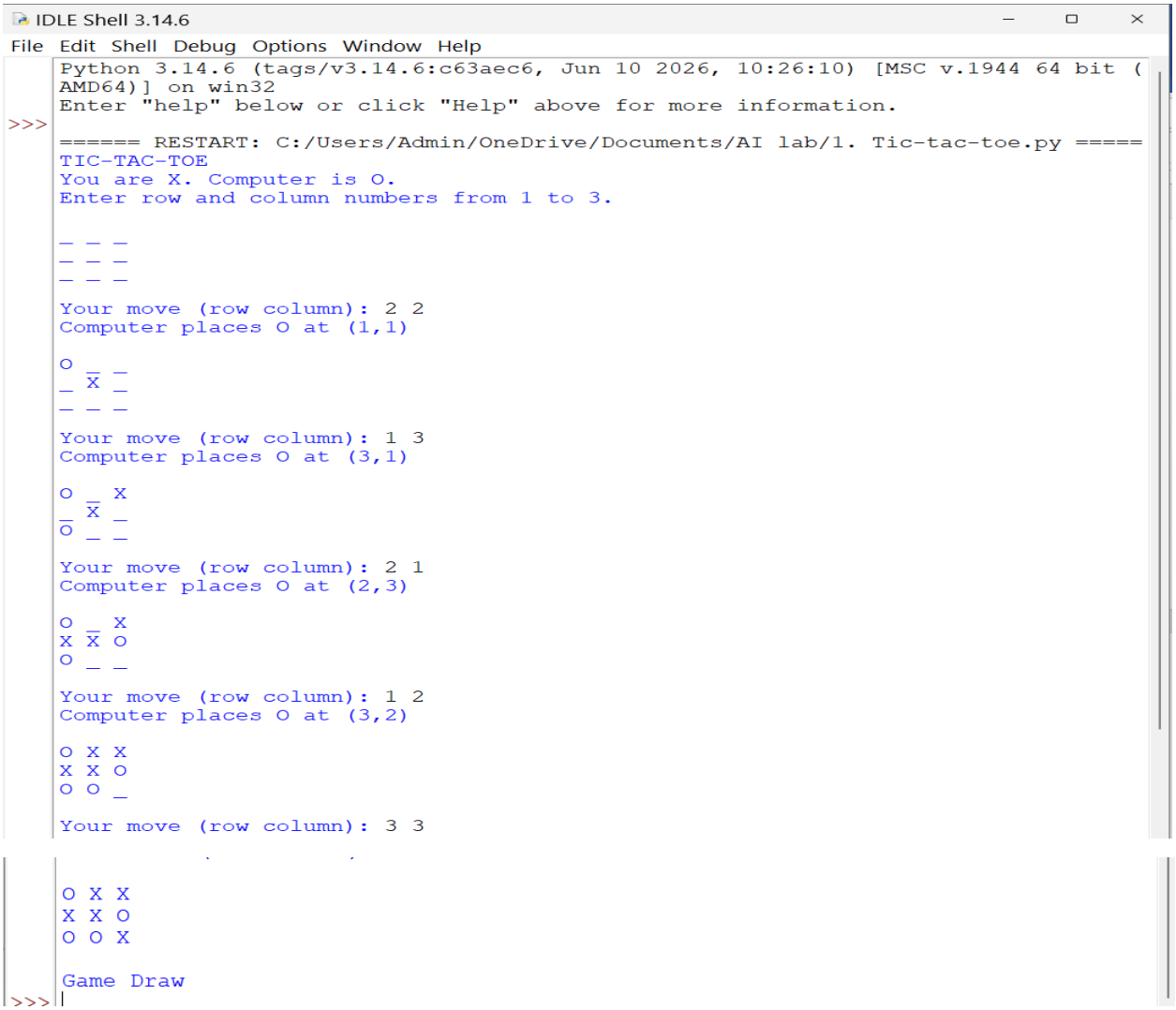
if __name__ == "__main__":
    main()
```

Output:

Test case 1: computer wins



Test case 2: Draw



Test case 3: Player wins

